

Enea Monzio Compagnoni

Final-year PhD in Computer Science, University of Basel · Zurich

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Summary

PhD researcher (ICML/ICLR/AISTATS) specializing in optimization for modern deep learning and reinforcement learning, with hands-on robotics experience building VLM/LLM-based agentic systems for autonomous robot operations (Flexion Robotics). I bridge theory and practice: I derive principled optimization insights, validate them with large-scale experiments (PyTorch/JAX). I am especially interested in natural-language-driven task planning/behavior orchestration, and integrating learned perception with planning for industrial robotics.

Eligibility & Availability

Available for a **6+ months full-time internship** in **2026**. Expected PhD graduation: **September 2026**.

Core Skills

Agentic AI & Planning: LLM/VLM-based agents, task decomposition and orchestration, prompting strategies, evaluation-driven iteration

Robotics & Evaluation: autonomous robot pipelines, sim-to-real reliability mindset, test methodologies, reproducible experiment setups

ML/RL: stochastic optimization, Transformers/LLMs, RL (PPO-style methods), differential privacy

Software: **Python** (JAX, PyTorch), clean research code, experiment tracking, integration/debugging, Git/Linux

Experience

Flexion Robotics AG

May 2025 – Aug 2025

AI Engineer — Robotics Intern (Perception / Agentic Systems)

- Implemented **VLM-based agentic frameworks** to enable autonomous robot operations.
- Developed prompting strategies; connected research ideas to practical, system-level robustness.
- Built evaluation environments and test methodologies, leading to reliable **outdoor demonstrations**
- Owned coordination and stakeholder alignment to deliver a **successful real-world demonstrations**.

University of Basel

Oct 2022 – Present

PhD Researcher & Lecturer, Computer Science

- Developed theory and actionable guidance for optimizer behavior via **SDE-based analyses** (SAM, AdamW, SignSGD, DCSGD), informing stability/scaling and hyperparameter schedules.
- Built **privacy-aware optimization** methods and training pipelines for **LLMs** under high-privacy regimes; validated results with large-scale experiments in Python/JAX.
- Current focus: improving policy optimization objectives and schedules (PPO-style methods) to make RL training more stable and reliable (relevant to robotics learning pipelines).
- Mentored PhD/MSc students; taught Deep Learning Foundations and Optimization.

Yahoo Research

Jul 2023 – Sep 2023

Intern, Scalable Machine Learning

- Built a risk-aware controller for ad-budget pacing; improved spend control by **~4%** in A/B tests and delivered production-adjacent Python components.

UBS AG

Jul 2019 – Sep 2022

AI Quantitative Analyst

- Built and improved ML systems for risk and forecasting in production settings (feature engineering, model iteration, stakeholder-facing delivery).
- Increased Japanese-market [liquidity-forecast](#) accuracy by $\sim 3\%$; supported AML detection improvements with new features and pipelines.

Selected Publications

- [E. Monzio Compagnoni](#), T. Liu, R. Islamov, F.N. Proske, A. Orvieto, A. Lucchi. Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise. ICLR, 2025.
- [E. Monzio Compagnoni](#), R. Islamov, F. N. Proske, A. Lucchi. Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs. AISTATS ([Oral](#)), 2025.
- [E. Monzio Compagnoni](#), A. Orvieto, H. Kersting, F.N. Proske, A. Lucchi. SDEs for Minimax Optimization. AISTATS, 2024.
- [E. Monzio Compagnoni](#), L. Biggio, A. Orvieto, F. N. Proske, H. Kersting, A. Lucchi. An SDE for Modeling SAM: Theory and Insights. ICML, 2023.

Full list: [Google Scholar](#).

Education

University of Basel

Oct 2022 – Sept 2026 (expected)

[PhD in Computer Science](#)

ETH Zürich

Sep 2018 – Apr 2022

[MSc in Quantitative Finance](#) (GPA 5.76/6, Summa Cum Laude)

University of Milan

2012 – 2018

[MSc & BSc in Mathematics](#) (110/110, Summa Cum Laude)

References

- Prof. Dr. Aurélien Lucchi (PhD Supervisor): aurelien.lucchi@unibas.ch
- Dr. Antonio Orvieto (External Collaborator): antonio@tue.ellis.eu

Languages

English, Italian